v0.3.3.1 SHVDN3 FINAL economy, pallets and wages review — EXPERIMENTAL

Issue #27 accepted-quote reload followup, October 22, 2026 (America/Chicago). The quote-persistence branch feature/shvdn3-repair-quote-reload is stacked on c05930f (manager wage action), pallet placement, #25 manual sales and #27 repair coverage from the PR #21 experimental SHVDN3 foundation. No branches were merged. This isolated followup closes the accepted-quote persistence gap found during final audit; the earlier guides/packages are historical snapshots and their stated reload gap is superseded by this candidate.

Final QA placement correction and review head

CURRENT final review head is experimental/shvdn3-economy-pallets-wages-review, a linear descendant of accepted-quote correction 604c622. No merge was performed. The final focused source delta is only TrySupplierPackageInteraction calling new RefreshStoragePalletAfterDeposit after the existing successful CommitSupplierDelivery. The helper resets the visual attempt budget and next-attempt time, then performs the ordinary dirty rebuild. No supplier accounting, inventory, schema, saved placement, quote, wage, sales or clock logic changes in this final delta.

A concrete final regression found that an exhausted five-attempt budget could skip repositioning a growing partial layer after a later successful deposit. Resetting the budget only at that explicit successful transition lets the original owned centered box move aside and prevents overlap with the new parcel. A failed/new model still has the existing bounded retry/rate limits: no per-frame requests or automatic infinite loop was added. Existing full stacks that cannot load after their ordinary attempts retain the documented stream-out/re-entry recovery. Two extra production-helper tests cover exhausted-budget partial-layer repositioning and bounded model requests without stock/invoice changes.

The final guide below describes the complete inherited requested stack. The current preservation.json/source.diff describe just this final placement delta against 604c622; complete-stack-source.diff includes all work since PR #21. Historical guides/results retain their own source hashes. The final Word/PDF report and validation.json are authoritative for this candidate.

Complete requested behavior

Repair coverage (#27, implemented first): fixed quote = max(existing service base plus existing expertise premium, estimated reserved parts acquisition plus allocated billable mechanic labor). Physical, background and remote paths share the calculation. Existing supplier procurement and payroll are not charged again. Universal/filter/brake/diagnostic parts use their actual dedicated-first/universal-fallback allocation, with two units for physical brake jobs. Quotes and source allocations stay fixed after acceptance, including when wages, equipment or configuration change later.

The inventory has counts, not purchase lots. Default unit estimates $125/$113/$200/$157 conservatively cover current express acquisition before discounts. Optional Settings.ini [Pricing] UnitCostEstimateUniversal, UnitCostEstimateFilters, UnitCostEstimateBrakes and UnitCostEstimateDiagnostics are nonnegative long estimates; missing/negative values select the source fallback and zero is deliberate. No historical acquisition cost or profit guarantee is invented. Labor uses mechanic wage/period and predicted billable repair duration, including existing wage bonus/specialty and speed effects. Walking, waiting, stalls, idle payroll, crash rework and later wage/config changes are outside that fixed billable estimate. Old promises remain old promises, even when below a newly configured estimate. RepairPricing.v1.csv is a separate completion audit; old customer CSV schemas are unchanged.

Manual parts sales (#25, implemented second): Business Management → Inventory & Suppliers → Parts Sales offers four stocked one-unit actions with quantity/availability/fixed-estimate preview. A fresh confirmation sells one unreserved selected unit, with shared input debouncing plus same-frame protection, no employee markup and no passive customer or mission. Pending deliveries/cargo and exact repair reservations are excluded. Cash/Revenue increase once; retail PartsSalesRevenue is separate from customer repair revenue and PartsSalesUnitsSold tracks

units. Both additive [Business] keys default to zero. A failed atomic save rolls back all sale changes. Supplier invoices, payroll and earned delivery-box visuals are untouched. PartsSales.v1.csv is separate.

Pallet placement (separate focused branch): up to nine earned boxes using cached model min/max bounds; prefer centered partial layers and 4–4–1, rotate with the saved heading and use the actual grounded pallet prop as world origin. Alternate fitting footprints and a 90-degree box turn are bounded fallbacks. Smaller footprints can make taller stacks; actual model/collision/MLO behavior and fallback realism await GTA testing. Invalid bounds fail placement before any new invoice/stock commit. Partial-layer rebuilds reposition owned props and a failed lower parcel hides unsupported upper owned visuals. Legacy mask gaps compact only visible rank from the bottom without inventing old boxes or changing saved masks/counts/capacity. Saved pallet positions and wholesale financial logic remain unchanged. Geometry is cached by at most eight model pairs and is not recalculated per frame.

Manager wages (separate focused branch): Business Management → Expenses & Bills → Pay Manager Wages shows the existing outstanding invoice and calls the same PayInvoice helper as Manager Profile. It pays separately from Pay All Bills, excludes the manager's own payroll rebate and preserves unrelated liabilities. No new wage schema or accrual was added. A failed individual payment save restores due, money, expenses, savings and invoice ages; repeat confirmations and both orders of individual/Pay All payment cannot pay the same invoice twice.

Accepted repair reload behavior

Before this followup: the foundation persisted financial totals but cancelled runtime customer/remote objects on reload; accepted quote history and exact allocation were lost. After: one additive bounded RepairPromisesV1 journal snapshots accepted quote data together with the ordinary business ledger. No GTA entity handle or world object is saved. Loading parses data only, retains a deliberately CLOSED shift, creates no entities, gives no income, and never uses file age or UTC elapsed to advance remaining work.

Physical accepted customers return through the existing configuration, proximity, reception, drop-off clearance, model loading and arrival path after the owner reopens the business. They keep order ID, price, cost breakdown, billable duration, exact parts allocation and last-saved work. They must reach the ordinary mechanic work/handoff phase before any payout. Current model/config selection supplies replacement entities, not stale handles. If staff changed, an available current mechanic can perform the existing promise at its fixed price/time. Entity-loss, streaming or close cancellation parks accepted physical work before resetting its worker timer; manager handoff keeps one remote identity rather than a second physical promise. Unaccepted random arrivals are not promised invoices and keep their existing cancellation behavior.

Background and remote promises restore only during positive active unpaused economy processing after reopening. An unavailable mechanic keeps work parked until an eligible current mechanic exists; remote promises wait for a manager. Existing mode rules still apply: background work requires background mode, physical customers require customer service/reception configuration, and remote work retains manager rules. Remaining time is the last saved active-work value; unsaved progress before a hard crash may need to be performed again. There is no offline catch-up. A ready-but-unpaid saved record still waits for a positive unpaused gameplay opportunity; physical work also waits for the normal NPC/vehicle phase.

Pending and respawned quotes reserve exact stock before their NPCs exist, so manual parts sales cannot drain it. Restoration does not consume or bill parts a second time, and handoff snapshots deduplicate the same order identity. Status lists parked order IDs, fixed amounts and remaining active seconds. No user data from older saves can reconstruct quotes that were never saved by any earlier build; those old saves default to an empty journal. The forward persistence guarantee starts with quotes accepted by this candidate.

Atomic settlement and failure behavior

TrySettleRepair centralizes all three repair payouts. A successful same-directory atomic business snapshot contains both the cleared promise and the one cash/revenue/stock/counter/XP update. On a failed snapshot it restores those fields, the physical Paid flag or remote queue position, and the mechanic's job state. Completion CSVs/notices are emitted only after success. Expense ledgers are unchanged. Physical assignment saves the accepted quote before vehicle tasks or an acceptance notice; if that save fails, assignment and premium are rolled back.

A failed repair commit retries at most once every three GTA game seconds per quote, using wrap-safe Game.GameTime arithmetic; it does not write every frame. Idle completed layouts do no placement calculations. Snapshot I/O is limited to existing save events, explicit actions and accepted/completed job transitions. The journal is bounded to 16 records; new acceptance stops before capacity would overflow. Closing the business parks all accepted routes before resetting workers, without advancing them. OnAborted snapshots promises before owned-entity cleanup, while preserving the original pre-activation abort guard. Startup delay/hold defaults and OnTick time sampling are unchanged.

RepairLedgerStamp binds a nonempty journal to the financial/inventory/job-counter values stored in the same snapshot. If an older build preserves unknown journal keys but changes cash, stock or completed jobs, returning to this candidate stops startup instead of replaying stale payable work. This is a consistency check, not an anti-cheat/authentication claim. Malformed/duplicate/nonfinite records, an unsupported version, too many records or a mismatching stamp throw before callbacks register and before any business save write. Preserve the file and request review or restore a matching full test backup; do not delete keys or silently discard invoices to bypass an error.

Schema, exact methods and before/after

No existing save/config value expression or SavePoint payload was changed in this followup. Save adds only WriteRepairPromises before its existing atomic commit. Old money, four stock buckets, expenses, supplier DeliveryCost, cargo, pallet coordinates and masks remain intact. Settings.ini gains nothing beyond the optional #27 pricing keys already described; no agent-edited save/settings is included.

Additive Save.ini sections: [RepairPromisesV1] Version=1, Count=0 by old-save default, LedgerStamp (empty when Count=0). [RepairPromiseV1.0] through .15 contain Route (0 background/1 physical/2 remote), MechanicSlot, OrderId, ServiceKind, DedicatedUnits, UniversalUnits, Baseline, Premium, PartsEstimate, LaborEstimate, Total, BillableSeconds, LegacyPromise and RemainingSeconds. Count is authoritative: stale unused sections are retained as unknown data but never replayed after Count clears. No timestamps drive work. The new source directly references one journal format/version; future formats require review.

Only experimental/shvdn3/Install/scripts/GlassHeroesBusiness.3.cs changes executable code. New data types/fields hold saved promises and retry state.
LoadRepairPromises/WriteRepairPromises/ValidRepairPromise/RepairLedgerStamp validate and serialize; CaptureRepairPromises/ParkRuntimeRepairPromises/RememberCancelledRepair preserve identity/progress; RestoreSavedNonphysicalRepairs/ApplySavedCustomerPromise/RepairQuoteHasStock restore the fixed work; TrySettleRepair commits or rolls back payment. Constructor, Save, ToggleBusinessShift, OnAborted, cancellation, customer dispatch/assignment/status, reservation and the three completion paths call them. The accepted-quote method boundary and focused diff are recorded in the historical saved-quotes guide/preservation report against c05930f. The CURRENT final placement delta is in this release's preservation.json/source.diff against 604c622; complete-stack-source.diff is relative to PR #21 430a202.

```
// Before: runtime objects cancelled on reload; completion mutated money then Save().
// After: data-only accepted promises load before callbacks, without world calls.
LoadRepairPromises(data);
// All repair routes use one durable terminal transition:
if (!TrySettleRepair(quote, mechanic, order, remote)) return;
// Save writes the original full payload, plus the bounded accepted ledger.
WriteRepairPromises(data);
```

The excerpt shows the rule; source.diff contains exact route-specific call sites, rollback fields and validation. Tests use extracted production helpers with a synthetic INI adapter and real temporary files; they do not certify the actual Windows ScriptSettings implementation, GTA entities or installed Enhanced runtime.

Reproduction and manual QA

From the full repository branch, with .NET SDK 8 available:

```
python verification/economy/verify_candidate.py --label final-review --profile final --suites
```

```
repair sales pallets wages reload
python verification/economy/package_candidate.py final-review --parent 604c622
```

The Release product is really compiled as net48/C#7.3 against official pinned ScriptHookVDotNet3 3.6.0. The source ZIP is separately extracted, all manifested files hashed and its linked product project rebuilt. Full protected-v2 scope verification requires the repository checkout; the ZIP deliberately contains no active v2 source. Its included verification files/logs support code review and behavior reproduction from the branch. The original v2 pallet test template is unchanged; its SHVDN3 compatibility template retains eight inherited invariants while replacing the obsolete guessed-height assertion with actual-bounds support.

Manual GTA QA (all NOT RUN): cold launch/reload past the 30-second default gate; open only deliberately. Accept all three physical services, including brake dedicated/fallback stock, then save/reload before assignment, during repair/vehicle handoff, at ready-unpaid and after payment. Confirm fixed quote, breakdown, allocation, remaining saved work and single payout; test changed wages/equipment/settings and temporarily absent staff/reception configuration. Repeat background and manager remote work, including no manager and mode disabled. Pause for minutes, close/reopen GTA, alter only synthetic file timestamps: no paused/offline earnings, wages or supplier catch-up may occur. Confirm pending repair stock stays unsellable while no NPC exists; excess stock can still sell once. Run interrupted-write/permission tests only in an isolated data copy, confirming unpaid/paid flag, stock, queue, XP and every total roll back with no repeated per-frame disk requests. Test deliberate older-build rollback on a COPIED save with pending quotes: modifying money/stock must produce a visible load error rather than duplicate revenue; retain the original intact backup.

Repeat four retail categories, held/mixed keyboard/gamepad confirmations, sold-out/reserved/incoming stock and save/reload after one sale. Verify existing supplier invoices remain owed until separately paid. Check all nine pallet deposits and partial layers at four cardinal headings and one arbitrary heading, elevated MLO ground, supported pallet variants, legacy mask holes, missing models and stream out/back. Inspect actual collision meshes and fallback height for floating/clipping/overhang/overlap. Compare saved coordinates/masks, stock, pending wholesale cargo and DeliveryCost before/after; only actual depot/pallet unloading may credit wholesale stock/invoice. Test Pay Manager alone, repeated/profile payment and both orders with Pay All, insufficient/exact funds, zero due, save failure and reload; preserve other bills and prevent self-rebate. Finish controlled heavy-mod/stress testing only after the owner's no-GH control and code review; preserve logs/dumps without attributing an unknown CLR overflow.

Installation, rollback and scope

This final ZIP is a complete review-only source package with exactly one .3.cs, the linked project/solution, current and historical focused guides/Word/PDF reports, source diffs, test/build logs and hashes. No runtime DLL, active v2 source, live Save.ini or Settings.ini is included. Do not copy the whole archive into GTA. Only after later owner authorization, back up the entire business data folder outside GTA, disable all duplicate/v2 Business Manager sources/DLLs, and copy only the intended experimental .3.cs. Keep installed SHVDN runtime files unchanged.

Rollback deserves care with nonempty journals: keep the source and its full matching save backup together. Older builds ignore accepted quote records and cannot resume them; if they change finances/stock, this candidate will detect a stale journal on return and stop rather than pay it twice. Prefer finishing pending work or using an isolated copied test save before testing older builds. Do not remove journal keys or mix financial snapshots; have ChatGPT review any migration needed. Empty journals remain ordinary additive-compatible saves. Restoring a pre-test full backup deliberately discards all later transactions.

No main, QA R2, development or PR #21 foundation merge/push was performed. No live GTA installation, business save or settings was read/edited by the agent. Manager supplier delivery automation, forensic branches and wholesale map blip remain separate. Installed Enhanced 3.9.0.6 compatibility NOT VERIFIED; Windows execution NOT RUN; GTA gameplay/stress testing NOT RUN. The 0xC00000FD cause remains UNATTRIBUTED / UNPROVEN, and these features are not evidence of a crash fix. GitHub draft creation is blocked by API Forbidden; see REVIEW_HANDOFF.md for pushed branches, exact PR bodies and comparison links.

Recorded automated results

Actual net48 / C# 7.3 Release build against ScriptHookVDotNet3 3.6.0: PASS, 0 errors, inherited warnings CS0169 and CS0649. SDK 8.0.426.

234/234 extracted-behavior tests PASS (85 inherited; repair: 32, sales: 29, pallets: 25, wages: 22, reload: 41). 21/21 static checks PASS. Source preservation/scope checks PASS; active-game OnTick clock sampling and original v2 source/project are unchanged. Tests use deterministic world stubs; they do not run GTA.

Source SHA256: 3146152304eb25599074e58403796719ced2eca9077eb003f917e81474d670f0. The complete unified source.diff applies to parent 604c622 and reproduces this source exactly.

GTA testing: NOT RUN. Windows execution: NOT RUN. Installed Enhanced 3.9.0.6 compatibility: NOT VERIFIED. Stack-overflow cause: UNATTRIBUTED / UNPROVEN. No installation, merge, protected-branch push, or live save/settings editing was performed.